Exhaustive AI-assisted review of research software finds merged, verified defects in AFNI and SPM, and a thirteen-year-old error in regional homogeneity changes a group result

Cynthia Condra
Unaffiliated `cindy.krafft@gmail.com`

September 2, 2026

Abstract

The software behind most published neuroimaging is large, old, mathematically dense, and read far less often than it is run. I asked whether an AI coding agent, directed to read an entire package adversarially and to verify every suspicion by executing code, could find defects that matter, and whether the maintainers would accept the fixes. For AFNI and SPM the answer to both is yes. Reading AFNI's nine numerical subsystems and all eleven of SPM's yielded 36 and 83 confirmed defects respectively, each with file-and-line evidence and a numerical reproduction; the highest-priority ones were sent upstream as ordinary pull requests, each with an issue carrying a runnable reproduction. Within a week the AFNI maintainers had merged two (tie handling in `3dReHo`; NIfTI descending slice timing) and the SPM maintainers two (artefact-window baseline offset in `@meeg/badsamples`; the sampling rate stamped by `spm_eeg_downsample`); further fixes are under review and are not reported here until the maintainers have adjudicated them. Only after the code review did I search the literature for exposure, which located five published ReHo studies in the affected regime, whose authors have been contacted, and cleared fifty-two others. To measure what one merged fix means for a published design I re-ran AFNI's standard resting-state pipeline on 40 participants of the open UCLA CNP dataset (20 controls, 20 with schizophrenia) and computed ReHo on identical residuals with the code immediately before and after the fix. Before the fix, `3dReHo` decided whether two samples were tied after truncating them to integers, so on percent-signal-change input it returned about 0.3 of the correct value and exactly zero in 17% of voxels; the two maps correlated only $r = 0.60$, and because the error scales with each participant's residual amplitude it does not cancel between groups: schizophrenia-versus-control t-maps from the two builds correlated 0.66, changed sign in a fifth of voxels, and their $p < 0.001$ sets did not overlap, although at this sample size no cluster survives family-wise correction under either build. The reviews, harnesses, patches, pipeline and results are public.

1 Introduction

Almost every quantitative result in neuroimaging passes through a few open-source packages that small teams have maintained for decades. The code is excellent and heavily used, and both facts work against complete review: a routine correct for the inputs its author had in mind can be wrong for inputs that became common later, and a test suite cannot cover a regime nobody thought to test. When such a defect surfaces it is treated as a singular event, as with cluster inference [1] or the platform-dependent chemical-shift scripts in organic chemistry [2], rather than as one instance of a class that could be searched for.

Large language models that can read and execute code make the search practical. This paper reports what happened when I pointed one at two of the most used neuroimaging packages, AFNI [3] and SPM, with a simple brief: read everything, assume nothing, and report only what you can reproduce. Two features of the design are deliberate. The review came first and the literature second, so that findings were not filtered by what I expected to matter. And nothing was reported to a maintainer without a numerical reproduction and, where possible, a patch, so that the unit of output is a pull request rather than a list of suspicions. Section 3 reports what was sent and what the maintainers merged. Section 4 then takes one merged fix and asks the question that matters to readers of the affected literature: how much did the results change?

2 Methods

2.1 Exhaustive adversarial review

An AI coding agent (Claude, Anthropic) worked in a sandboxed environment with the package source checked out, a compiler, and the ability to run code. For each package the source was divided into subsystems and each subsystem was read in full with an adversarial brief: for every routine, what inputs are legal but unanticipated, what happens at boundaries and at zero, where do numeric types change, where is a convention assumed rather than checked. AFNI (`afni/afni` master at `0488c9e`, 27 August 2026) was divided into nine subsystems where a correctness bug most directly changes a published number: the statistical distribution library, group statistics in C, cluster-extent inference, the single-subject GLM, registration and NIfTI I/O, resting-state and connectivity tools, the R-based group programs, dataset arithmetic, and the Python pipeline with DTI/PCA tools. SPM (`spm/spm` development version at `530ec52`, 23 August 2026) was divided into eleven: statistical distributions, random field theory and results, the GLM core, spatial preprocessing, file I/O, the C/MEX sources, DCM and Bayesian inversion, numeric and mesh utilities, M/EEG processing, the spatial toolboxes, and the active-inference code. A human directed the agent at the level of package, subsystem and which candidates to pursue; the agent wrote the reviews, harnesses and patches. Every candidate was recorded with file and line evidence.

2.2 Verification by execution

No candidate was reported without one of three kinds of execution: compiling the routine under test against an independent or high-precision reference; porting it faithfully, idiosyncrasies included, and testing the port against ground truth on synthetic data; or running the shipped program and observing the defect. In AFNI this meant standalone C harnesses compiled from the actual translation units, comparison against SciPy reference distributions, fuzz tests against sorted references and Monte-Carlo simulation. In SPM it meant Python transcriptions checked against NumPy and SciPy or closed forms, a Monte-Carlo simulation of smooth χ^2 random fields, and, for the fix to `spm_ECdensity`, executing the real MATLAB code in GNU Octave before and after the patch with a new unit test (12 of 23 assertions fail before, 0 after). Candidates that did not survive were recorded as such.

2.3 Upstream filing

Findings that survived were ranked by severity times how often the code path runs, and the top of each list was filed in the form each project uses: GitHub issues with runnable reproductions and pull requests carrying the patch and the harness. The maintainers' review is the adjudication.

2.4 Literature exposure, assessed afterwards

Exposure was assessed only after the review, and only for the findings at the top of each list, by reading methods sections for the specific trigger of each defect. For the `3dReHo` finding this meant a full-text search

Table 1: The 40 participants. Motion is the mean Euclidean norm of the frame-to-frame motion derivative (mm), over all volumes and over the volumes retained after censoring. Group comparisons: Student t (df 38) or Fisher's exact test.

	CONTROL ($n = 20$)	SCHZ ($n = 20$)	p
Age, years (range)	32.4 ± 8.6 (21–49)	36.0 ± 9.7 (22–49)	0.21
Sex, F / M	8 / 12	6 / 14	0.74
Motion, all volumes	0.103 ± 0.036	0.123 ± 0.045	0.11
Motion, retained volumes	0.093 ± 0.028	0.105 ± 0.034	0.24
Volumes censored	5.0%	8.8%	0.10
Residual SD handed to `3dReHo`	0.370 ± 0.057	0.334 ± 0.082	0.11

of Europe PMC for `3dReHo` and for ReHo used with AFNI (398 unique records, 310 full texts retrieved) plus every work citing the tool's description [4], giving 57 papers that computed ReHo with AFNI, each adjudicated on one question: in what units was the file passed to `3dReHo`? For SPM the same after-the-fact reading anchored each Tier-1 finding to the methods papers that document the affected machinery. A separate six-journal full-text survey (44,198 articles, 2021–2026; 20,501 readable) served as a denominator for how often each package is used, not as the selector of what to read.

2.5 Reanalysis of the ReHo defect on open data

Data and pipeline. Resting-state fMRI and T1-weighted images from the UCLA Consortium for Neuropsychiatric Phenomics dataset, OpenNeuro ds000030 [5] (3 T, TR 2 s, 152 volumes). I attempted the first 20 eligible CONTROL and 20 SCHZ participants by identifier and drew from a spare pool until 20 of each passed `afni_proc.py` (AFNI 26.2.06; blocks despike, tshift, align, tlrc, volreg, blur 4 mm, mask, **scale**, regress; MNI152 2009 template; band-pass 0.01–0.1 Hz; censoring at 0.3 mm and 5% outliers; motion parameters and derivatives as regressors). The default `scale` block converts the data to percent signal change, so the residual series handed to `3dReHo` have SD of order 0.3–0.5. Twelve of 52 attempted participants (4 of 24 CONTROL, 8 of 28 SCHZ) failed the pipeline's own regression step because censoring left fewer time points than the 108 nuisance regressors and were replaced; that exclusion is identical for both arms. The two groups did not differ significantly in age, sex, head motion or censoring (Table 1), although the SCHZ group moved more and lost more volumes, as clinical groups typically do.

The two builds. The fix landed in two commits on consecutive days: the tie-detection change itself (`94ee52b`, pull request 944) and a guard returning $W = 0$ when the tie correction exactly cancels the denominator (`74a90ac`). To isolate the tie-detection change I compare the source immediately before it (`4c2bd54`) with the guard applied against master at `d202535` (AFNI 26.2.06, bit-identical to the distributed build on test data), so the only difference is whether ties are detected on truncated integers or on floating-point values. `3dReHo -nneigh 27` from each build was run on the identical residuals and mask of every participant. A faithful Python port reproduces both builds voxel for voxel to four decimals on synthetic volumes. Releases before 26.2.04 (August 2026) did not close a tie run reaching the end of the sorted series, so in those releases a voxel whose entire series lies inside one integer bin is returned uncorrected, hence correct, rather than as zero; the truncation defect itself is common to every release.

Analysis. Per participant: the ratio and relative error of before-fix to after-fix ReHo over all in-mask voxels, the fraction of voxels the before-fix build returns as exactly zero, and the voxelwise correlation of the two maps. Group level: `3dttest++` two-sample SCHZ versus CONTROL on each set of maps, in the intersection of all 40 masks (8,657 voxels) and in voxels present in at least 90% of participants (25,227

Table 2: Correctness fixes from the AFNI and SPM reviews merged upstream, as of 2 September 2026. Each pull request is paired with an issue carrying a runnable reproduction.

Package	PR	Change	Merged
AFNI	944	`3dReHo`: detect rank ties on the float values, not integer-truncated copies; on percent-signal or standardised input almost every sample had been declared tied and Kendall's W collapsed (Section 4)	28 Aug 2026
AFNI	947	NIfTI reader: loop bound for sequential-descending slice timing compared against the wrong end, so every slice offset was zero and `3dTshift` silently did nothing, including on files AFNI itself writes	28 Aug 2026
SPM	163	`@meeg/badsamples`: peristimulus baseline added a second time when mapping artefact events to samples, shifting every bad-sample window on epoched data by the baseline length	2 Sep 2026
SPM	165	`spm_eeg_downsample`: stamp the achieved, not the requested, sampling rate (1000→180 Hz actually yields 166.7 Hz)	2 Sep 2026

voxels); the `3dttest++` output was reproduced to the last digit by an independent t-test. Voxelwise maps were compared at uncorrected thresholds, since the question is how the two builds' maps differ rather than whether either is significant. Cluster-level inference, as a published design would report it, used `3dttest++ -Clustsim` (sign-flip randomisation of the residuals, 10,000 iterations, first-nearest-neighbour clustering, bi-sided) to obtain the cluster-size threshold at family-wise $\alpha = 0.05$ for cluster-forming $p = 0.001$ and $p = 0.01$ under each build and mask, followed by `3dClusterize`. No covariates were modelled; the groups did not differ significantly on age, sex or motion (Table 1).

3 Results: what was found and what was merged

Table 2 lists the correctness fixes the maintainers have accepted. Further pull requests and issues are open on both projects; they are deliberately not described here, so that the maintainers can adjudicate them before anything is claimed. Behind them stand 36 confirmed findings in AFNI (9 judged capable of changing published numbers, 15 narrower, 12 likely) and 83 in SPM (5 high, 15 medium, 63 low priority), each with file-and-line evidence and a harness in the repository, alongside the code paths that were verified correct, which are recorded with equal care: in SPM the standard fMRI GLM was traced end to end (`spm_spm`, the ReML machinery, the T and F Euler-characteristic densities, both FDR procedures, the NIfTI quaternion core) and held; in AFNI more than forty high-stakes paths held, most verified numerically.

Two things about the merged set are worth stating. None was a suspicion: each pull request carried a reproduction the maintainer could run, and in each case the maintainer's comment acknowledged the mechanism (for PR 944, that a mapping from values to ranks "had not been added"; for PR 947, that descending sequences "see minimal real-world usage" despite affecting AFNI's own output). And none would have been found by starting from the literature: no methods section says "we passed percent-signal data to a routine that truncates to integers." The review found the defect; the literature search, done afterwards, found who was exposed.

3.1 Exposure, found after the fact

For the `3dReHo` defect, 57 papers were adjudicated. Five are potentially affected: two normalised each voxel's series to unit variance, the worst case, and three ran `afni_proc.py` with its default scaling. I do not identify them here. Whether a defect in the software changes a study's conclusions is a question only a rerun by its authors can answer, and the corresponding authors of the five studies have been contacted with the finding, the fix and the reanalysis below. Fifty-two are not, most of them cleared at once by the scaling conventions of the large wrapper pipelines (XCP-D keeps native units; HALFpipe, PhiPipe and CCS grand-mean scale to 10,000; C-PAC and DPABI do not call `3dReHo`). One paper that L2-normalised its input was first classified as exposed and moved out when the faithful port showed that regime to be exactly correct; that reversal is recorded because it is what verification is for. Eleven papers could not be read. For the SPM findings the exposure is anchored to methods papers rather than to individual studies: the artefact-window defect sits in the machinery documented by the SPM8 M/EEG reference paper, and the free-energy defect in DCM for evoked responses as introduced in its methods papers; a per-paper join for SPM, in the style of the ReHo pass, is future work.

4 Results: one merged fix, measured end to end

4.1 The defect

Regional homogeneity [6] is Kendall's coefficient of concordance W over the ranked time series of a voxel and its neighbours, with a correction $\sum_i (t_i^3 - t_i)$ over tie runs of length t_i subtracted from the denominator. In AFNI's implementation [4] (`src/ptaylor/rsfc.c`, lines 63–118) ranks came from a true floating-point sort, but tie *detection* compared values stored into an integer array, so any two samples sharing an integer part were declared tied. For raw scanner units this almost never happens. For percent-signal-change residuals or z-scored series, nearly every sample lies in $(-1, 1)$ and truncates to zero: almost the whole series becomes one tie run, the correction approaches the denominator, and W collapses toward zero, reaching exactly zero in voxels whose whole series lies in one integer bin. Figure 1 shows the behaviour of the faithful port against input scale with the 40 participants overlaid. The error is not a constant factor: it depends on the amplitude of each voxel's own series, so the before-fix statistic is partly a measure of BOLD variance; and the affected regime is exactly where `afni_proc.py`'s default `scale` block and every "normalized to unit variance" step put the data.

4.2 Per participant

Across the 40 participants the residual SD handed to `3dReHo` averaged 0.35 (range 0.14–0.52). The before-fix build returned on average 0.28 of the after-fix whole-brain mean ReHo (0.134 versus 0.481), a relative error over all voxels of 75% (range 61–92%); it returned exactly zero in a mean 17% of in-mask voxels (range 2–69%) and about 0.31 of the correct value elsewhere. The error tracks residual amplitude across participants ($r = -0.89$). The before- and after-fix maps correlated $r = 0.60$ across voxels (range 0.31–0.76), the correlation rising with residual SD ($r = 0.68$; Fig. 2B). The defect reorders voxels rather than rescaling them, to a degree that varies with each participant's signal amplitude.

4.3 Between groups

Residual amplitude and censoring differed between the clinical groups (SCHZ residual SD 0.334 ± 0.082 versus CONTROL 0.370 ± 0.057; zeroed voxels 20% versus 14%; censored volumes 8.8% versus 5.0%), so the bias is group-dependent. Table 4 and Figure 2C give the contrasts. The t-maps from the two builds correlated 0.66–0.68, disagreed in sign in 18–21% of voxels, and their $p < 0.001$ sets did not overlap (0 of

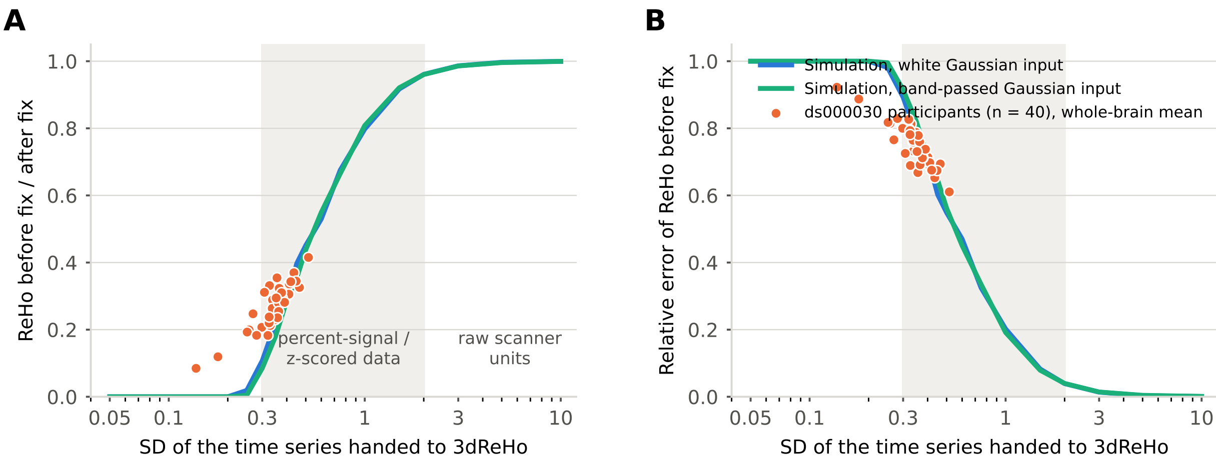

Figure 1: The defect as a function of input scale. **A**, Ratio of before-fix to after-fix ReHo against the SD of the time series handed to `3dReHo`, from a faithful port of the routine on Gaussian input (blue: white; green: band-passed 0.01–0.1 Hz at TR 2 s), 27-voxel neighbourhoods, 148 time points, 60 simulations per point. Orange: the whole-brain mean ratio for each of the 40 ds000030 participants at the SD of their own residuals. **B**, The same as relative error. Shaded band: percent-signal-change and z-scored data. Raw scanner units (SD $\gtrsim 5$) are unaffected.

Table 3: Cluster-level inference for SCHZ versus CONTROL under each build (`3dttest++ -Clustsim`, 10,000 sign-flip randomisations, NN1, bi-sided, $\alpha = 0.05$).

Mask	Cluster-forming p	Build	Size threshold (vox)	Largest cluster found (vox)
90%-coverage	0.001	before / after	24 / 24	21 / 20
	0.01	before / after	151 / 179	114 / 138
Intersection	0.001	before / after	14 / 16	0 / 3
	0.01	before / after	93 / 88	31 / 57

12 in the intersection mask; 2 of 104 in the 90%-coverage mask); even at $p < 0.05$ the two builds agreed on a quarter of their union (Jaccard 0.24 and 0.29). The whole-brain mean reduction of ReHo in schizophrenia, the simplest statistic such papers report, was $t(38) = -2.25$, $p = 0.031$, $d = -0.71$ after the fix and $t = -1.74$, $p = 0.090$, $d = -0.55$ before it.

4.4 Cluster-level inference

Table 3 reports the analysis a published design would: family-wise cluster correction at $\alpha = 0.05$. Under neither build does any cluster survive, at either cluster-forming threshold, in either mask. In the 90%-coverage mask the largest $p < 0.001$ clusters are 21 voxels before the fix and 20 after, against a threshold of 24; the largest $p < 0.01$ clusters are 114 and 138 against 151 and 179. Twenty participants per group is underpowered for a corrected whole-brain ReHo difference between schizophrenia and control under either build, so in this pilot the fix's effect is on the content of the maps, which the uncorrected comparisons above describe, and not on whether a corrected result exists. This is itself informative: the exposed studies report corrected clusters from comparable or smaller samples, and clusters near a threshold are exactly where a map-reordering error of this size would move a result across it.

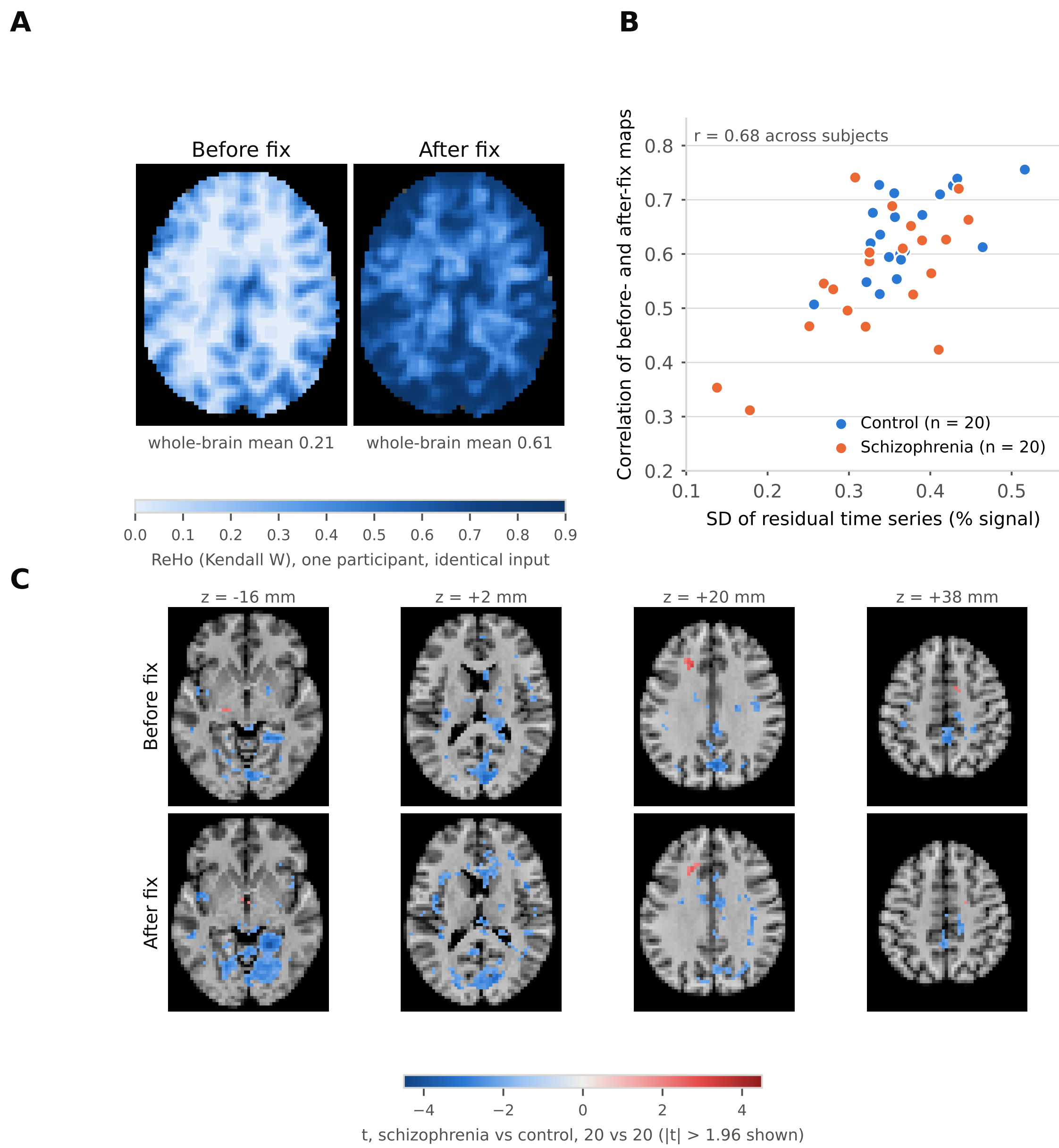

Figure 2: One defect, before and after, on identical data. A, ReHo for one control participant computed from the same residual time series by the before-fix `3dReHo` (left) and the after-fix build (right), same slice and colour scale. The before-fix values are about one third of the correct ones, exactly zero in 7.5% of this participant's mask (the palest regions), and differently arranged. **B**, For each of 40 participants, the voxelwise correlation between before- and after-fix maps against the SD of that participant's residuals. **C**, SCHZ-versus-CONTROL t-maps (20 vs 20) from the before- and after-fix maps, $|t| > 1.96$ shown, four axial slices, 90%-coverage mask.

Table 4: SCHZ versus CONTROL (20 vs 20, `3dttest++`), same participants and residuals, the only difference being which `3dReHo` produced the input maps.

	Intersection mask (8,657 vox)	90%-coverage mask (25,227 vox)
Voxels $\lvert t\rvert > 3.57$ ($p < .001$), before / after	0 / 12	49 / 57
overlap (before $\cap$ after / before $\cup$ after)	0 / 12	2 / 104
Voxels $\lvert t\rvert > 1.96$ ($p < .05$), before / after	767 / 920	2,849 / 4,470
overlap (Jaccard)	324 / 1,363 (0.24)	1,632 / 5,687 (0.29)
Spatial correlation of t-maps	0.66	0.68
Voxels where t changes sign	21%	18%
Mean $\lvert t_{\text{before}} - t_{\text{after}}\rvert$	0.66	0.69

5 Discussion

What the case study establishes. Every ReHo map computed by AFNI on percent-signal-change or standardised input before 27 August 2026 was affected, and not in the benign way of a wrong scale factor. In a typical participant the before-fix map was a third of the correct value with a different spatial arrangement and a sixth of the brain at zero; between groups whose signal amplitude differs, as clinical groups typically do, the before-fix statistic mixes a synchrony difference with an amplitude difference, and the voxels that reach significance are largely different ones. I make no claim about the conclusions of the five potentially affected studies, which is why they are not named; none shares its data, and only reruns by their authors, who have been contacted, can settle what changes. The reanalysis is of their design, on open data, with nothing but the fix varied, and it says the reruns are worth doing. Its limits are those of a pilot: 40 participants, one dataset, one configuration, no covariates, a $p = 0.05$ crossing on the global mean that is illustrative rather than evidential, and no cluster surviving family-wise correction under either build, so that the between-build comparison rests on uncorrected maps by design. It isolates the tie-detection change given the concurrent divide-by-zero guard; earlier releases left the lowest-amplitude voxels uncorrected rather than zeroed, which changes the size of the error in those voxels but not the mechanism or the amplitude dependence that drives the group-level result.

Why review-first matters. Each of the four merged defects was invisible from the literature side. A methods section does not say which loop bound the NIfTI reader compared against, whether the artefact detector and the artefact reader agreed about the baseline, or whether a sampling rate was requested or achieved. Starting from the papers, as reproducibility efforts usually do, would have found none of them. Starting from the code found all four, and the literature pass afterwards found who was exposed, including a reversal (the L2-normalised regime) that only the faithful reproduction could have produced.

Why the defects survived. Each routine was correct for the data its author tested it on; the regime in which it fails was created later, by another program's default, by a rarely used slice order, by an epoching convention, by a non-integer ratio. No unit test exercised the regime (`spm_ECdensity` had none at all) and no user could see a defect whose signature is "lower ReHo everywhere, more so in noisier participants." This is the general shape of the findings: not carelessness but the combinatorics of large, old codebases and evolving conventions, which no team can read exhaustively with the question "what if the input is small, or tied, or descending" in mind for every line.

What machine-assisted review changes. That question is now cheap to ask of every line. Two features of the method are essential and I would caution against dropping either. Verification by execution is what separates a finding from a plausible story; a substantial fraction of the agent's candidates did not survive it. And upstream review is the adjudication: the merged fixes are the deliverable, and the maintainers' reading of a patch, not ours, is what makes it trustworthy. Given both, the economics change: an exhaustive adversarial read of AFNI or SPM, with a numerical check of each suspicion, costs days of compute rather than years of expert time, and its output arrives in the form maintainers already accept.

Recommendations. Funders and journals should treat audits of the software behind published results as reproducibility infrastructure rather than as one-off events. Maintainers should be resourced to receive and review such reports, because a verified finding has value only once it is a merged fix and a corrected literature. And authors should publish their analysis scripts together with the exact versions of every piece of software they ran. The 57-paper exposure pass here took days of reading and still left eleven papers unresolved, because methods sections say "ReHo was computed in AFNI" and not which build, in which units, with which options; a published script and version string answer all three at once. Software will keep having bugs. Whether the literature can be corrected when one is found depends on whether it is possible to tell, quickly and mechanically, who ran the affected code, and today it mostly is not.

Limitations. "Confirmed" means traced end to end and reproduced as stated, not adjudicated by the maintainers; where they disagree, their reading wins. The SPM Tier-1 fixes other than `spm_ECdensity` were verified by transcription and reasoning rather than by executing MATLAB, which is stated in each pull request. Exposure was assessed only for the top findings, and for SPM only at the level of methods papers. The literature pass reaches only openly readable full text, so a paywalled paper that says "ReHo was computed in AFNI" is invisible to it. The same repository holds reviews of seven further packages at varying stages of verification and filing; they are not reported here because their fixes have not yet been through maintainer review.

Data and code availability

The component reviews, every verification harness, the patches and issue texts, the reanalysis pipeline (`run_subject.sh`, `group_analysis.sh`, `subject_stats_v2.py`), subject lists, per-participant tables, group t-maps and figures are at `https://github.com/cindykrafft/research-software-audit`. Imaging data are OpenNeuro ds000030. The `3dReHo` builds are reproducible from `afni/afni` at `4c2bd54` (plus `74a90ac`) and `d202535`. Merged pull requests: `https://github.com/afni/afni/pull/944`, `https://github.com/afni/afni/pull/947`, `https://github.com/spm/spm/pull/163`, `https://github.com/spm/spm/pull/165`.

Disclosure of AI use

The code reviews, harnesses and patches described here were produced by an AI coding agent (Claude, Anthropic) operating under the author's direction; every reported finding was verified by executing code, and every upstream filing was reviewed by the author before submission. The same system assisted in drafting this manuscript and produced its figures from the checked-in results. The author has no financial interest in any software named.

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